

Lekhapriya Dheeraj Kashyap

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Research Interest

Human-Centered AI, Computational Behavior Modeling, AI-assisted Decision Systems, Partially Observable Decision Process, Stochastic Modeling, Inverse Learning, Structural Estimation and Optimization

Education

Texas A&M University PhD in Industrial & Systems Engineering	College Station, TX 2021 – 2026
New Jersey Institute of Technology M.S. in Industrial & Engineering Management	Newark, NJ 2014 – 2016
Visvesvaraya Technological University B.Eng. in Electronics and Communication	Bangalore, India 2010 – 2014

Research Experience

Southern Methodist University <i>Postdoctoral Researcher</i> <i>Advisor: Prof. Yanling Chang</i>	Dallas, TX 2026 - present
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- Modeled commercial fishing as a continuous-state Partially Observable Markov Decision Process, linking biological stock recruitment dynamics to a dynamic discrete-choice model of fishers' daily go/no-go decisions. Developed a particle filter belief projection method to make continuous-state POMDP estimation computationally tractable, and applied two-stage maximum likelihood to jointly recover catchability and fishing costs to enable counterfactual simulation of fishery management policies.

Texas A&M University <i>Research Assistant</i> <i>Advisors: Prof. Alfredo Garcia & Prof. Yanling Chang</i>	College Station, TX 2021 - 2026
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- Developed deep learning methods to learn compact latent representations from high-dimensional, multi-modal human behavioral data. Jointly inferred system dynamics, agent preferences, and utility functions from expert demonstrations to support interpretable human–AI decision modeling.
- Formulated driver attention and multitasking as a Partially Observable Semi-Markov Decision Process and applied structural inverse reinforcement learning to infer latent distraction states, belief dynamics, and infer reward functions from behavioral data. Enabled personalized modeling and evaluation of in-vehicle human–AI interaction and intervention strategies.
- Developed probabilistic latent-state models to infer unobserved context, attentional effort, and reward trade-offs from large-scale, naturalistic human decision data. Applied Bayesian inference in partially observable settings to support interpretable evaluation of human performance, bias, and AI-assisted decision policies.
- Applied multi-agent reinforcement learning to decentralized power systems, addressing coordination, scalability, and stochastic optimization under uncertainty while preserving local operational autonomy.

Texas A&M University

Research Assistant

Advisors: Prof. Satish Bukkapatnam & Prof. Ceyhun Eksin

College Station, TX

2023 - 2024

- Developed an actor-critic (PPO) framework for constrained decision-making in multi-stage manufacturing. Integrated stochastic digital twins to enable adaptive control and joint optimization of process parameters to achieve precise surface roughness targets while minimizing energy and operational costs.

Indian Institute of Science

Research Associate, Department of Management Studies

Advisor: Prof. Shashi Jain

Bangalore, India

2019 - 2021

- Derived unbiased stochastic gradient estimators for maximum likelihood optimization of multi-dimensional Hawkes processes with neural network parameterization of excitation kernels. Validated the approach on real-world finance data, demonstrating strong performance in high-frequency event modeling.

Publications (Published)

- **Lekhapriya Dheeraj Kashyap**, Zhide Wang, Yanling Chang, Alfredo Garcia. “A Structural Model of Attentional Effort Dynamics: Evidence From a Naturalistic Discrimination Task”. *Human Factors*, 2025. [Paper](#);
Spotlight: Human Factor Journal’s Editor’s Pick, 2025, [News](#);
Runner up award: Poster Presentation, Texas A&M University, 2024,
- **Lekhapriya Dheeraj Kashyap**, Zhide Wang, Yanling Chang, Maryam Zahabi, Alfredo Garcia. “Inferring Hidden Attentional States in Driving: A Bayesian Approach to Modeling Distraction and Secondary Task Engagement”. *Human Factors*, 2025. [Paper](#);
Spotlight: Texas A&M University, College of Engineering, 2026, [News](#);
- Avisek Naug, Antonio Guillen, Ricardo Luna, Vineet Gundecha, Cullen Bash, Sahand Ghorbanpour, Sajad Mousavi, Ashwin R Babu, Dejan Markovikj, **Lekhapriya Dheeraj Kashyap**, Desik Rengaranjan, Soumyendu Sarkar. “Sustaindc: Benchmarking for sustainable data center control”. *Advances in Neural Information Processing Systems*, 2024. [Paper](#)
- Avisek Naug, Antonio Guillen, Ricardo Luna Gutiérrez, Vineet Gundecha, Sahand Ghorbanpour, **Lekhapriya Dheeraj Kashyap**, Dejan Markovikj, Lorenz Krause, Sajad Mousavi, Ashwin Ramesh Babu, Soumyendu Sarkar “Pydcn: Custom data center models with reinforcement learning for sustainability”. *Proceedings of the 10th ACM International Conference on Systems for Energy-Efficient Buildings, Cities, and Transportation*, 2023. [Paper](#)
- Sobin Joseph, **Lekhapriya Dheeraj Kashyap**, Shashi Jain. “Shallow Neural Hawkes: Non-parametric kernel estimation for Hawkes processes”. *Journal of Computational Science*, 2022. [Paper](#)

Publications (Manuscripts Under Review)

- Adithyaa Karthikeyan, **Lekhapriya Dheeraj Kashyap**, Ceyhun Eksin, Satish Bukkapatnam. “Actor-critic based reinforcement learning for multi-stage process planning and end point control in abrasive finishing operations”

Publications (Manuscripts in Preparation)

- **Lekhapriya Dheeraj Kashyap**, Yanling Chang, Maryam Zahabi, Alfredo Garcia. “State Representation and Preference Learning from Demonstrations of a Distracted Driver” ([Github](#))

- **Lekhapriya Dheeraj Kashyap**, Yanling Chang, Zhide Wang, Matthew Keblis “Fishing Decisions under Uncertain Crab Stock”
- **Lekhapriya Dheeraj Kashyap**, Alfredo Garcia. “Multi-Agent Reinforcement Learning for Decentralized Power Systems” ([Github](#))

Industry Experience

Hewlett Packard Enterprise

Research Associate Intern, AI-RL Lab

Milpitas, CA

2023

- Developed configurable benchmarking environments for multi-agent reinforcement learning in data center operations, modeling coordinated workload scheduling, cooling, and energy storage under shared constraints. Evaluated MARL algorithms to analyze system behavior, energy efficiency, and carbon-aware control strategies for real-world infrastructure.
- Built a high-performance, configurable data center simulator with a Gymnasium-compatible interface, enabling rapid prototyping and evaluation of reinforcement learning-based cooling strategies for sustainable system design.

WellsFargo

Quantitative Intern, Market & Counterparty Risk Analytics

Charlotte, NC

2022

- Leveraged Bloomberg BVAL sector- and rating-level time series to model bond portfolio risk. Implemented Python-based risk engines to compute and assess Value at Risk (VaR) and SVaR. Built SQL pipelines for daily P&L computation and performed P&L attribution and backtesting against BVAL-derived risk factors. Analyzed and reported key drivers of material deviations relative to production benchmarks to senior management.

Teaching Experience

Lab Instructor

Course: Quality Engineering (ISEN 350)

Industrial & Systems Engineering, Texas A&M University

Fall 2021, Spring 2022, Fall 2022,

Spring 2023, Fall 2025

- Led a weekly lab session for a class of undergraduate students
- Graded lab reports for students
- Interacted with students during weekly office hours
- Student evaluations across five semesters were consistently positive, with an overall response-weighted average of 4.49/5 for fostering an effective learning environment, 2.69/3 for teaching methods contributing to learning, and 2.65/3 for encouraging students to take responsibility for their learning.

Teaching Assistant

Course: Uncertainty Modeling for Industrial Engineering (ISEN 310)

Industrial & Systems Engineering, Texas A&M University

Fall 2023

- Graded assignments and tests for undergraduate students
- Interacted with students during weekly office hours

Teaching Assistant

Course: Decision and Risk Analysis (ISEN 347)

Industrial & Systems Engineering, Texas A&M University

Spring 2025

- Graded assignments and tests for undergraduate students

Presentations and Posters

- Guest speaker for Engineering Graduate Seminar Series, titled “Modeling and inferring latent attention in partially observable decision making systems”, *LeTourneau University, 2026*
- Guest lecture for DAEN/ISEN 489, “Introduction to Reinforcement Learning and Partial Observability”, *Texas A&M University, 2025*
- **Lekhapriya Dheeraj Kashyap**, Zhide Wang, Yanling Chang, Alfredo Garcia. “Attention Allocation is Adapted to Context: Evidence from a Naturalistic Discrimination Task”, *Zorich’s Reliability Workshop, Department of Statistics, Texas A&M University, 2024*

Honors and Awards

- Second place, Poster Presentation, Zorich’s Reliability Workshop, Department of Statistics, Texas A&M University, 2024
- Runner up, paper title: “Reinforcement Learning based End Point Control for Polishing Operations”, IISE Annual Conference and Expo, 2024
- Paper highlighted as Human Factor Journal’s Editor’s Pick, 2025, [News](#)

References

Dr.Alfredo Garcia

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